

# Lumina

Crime Intelligence & Analytical Platform

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## System Design Documentation

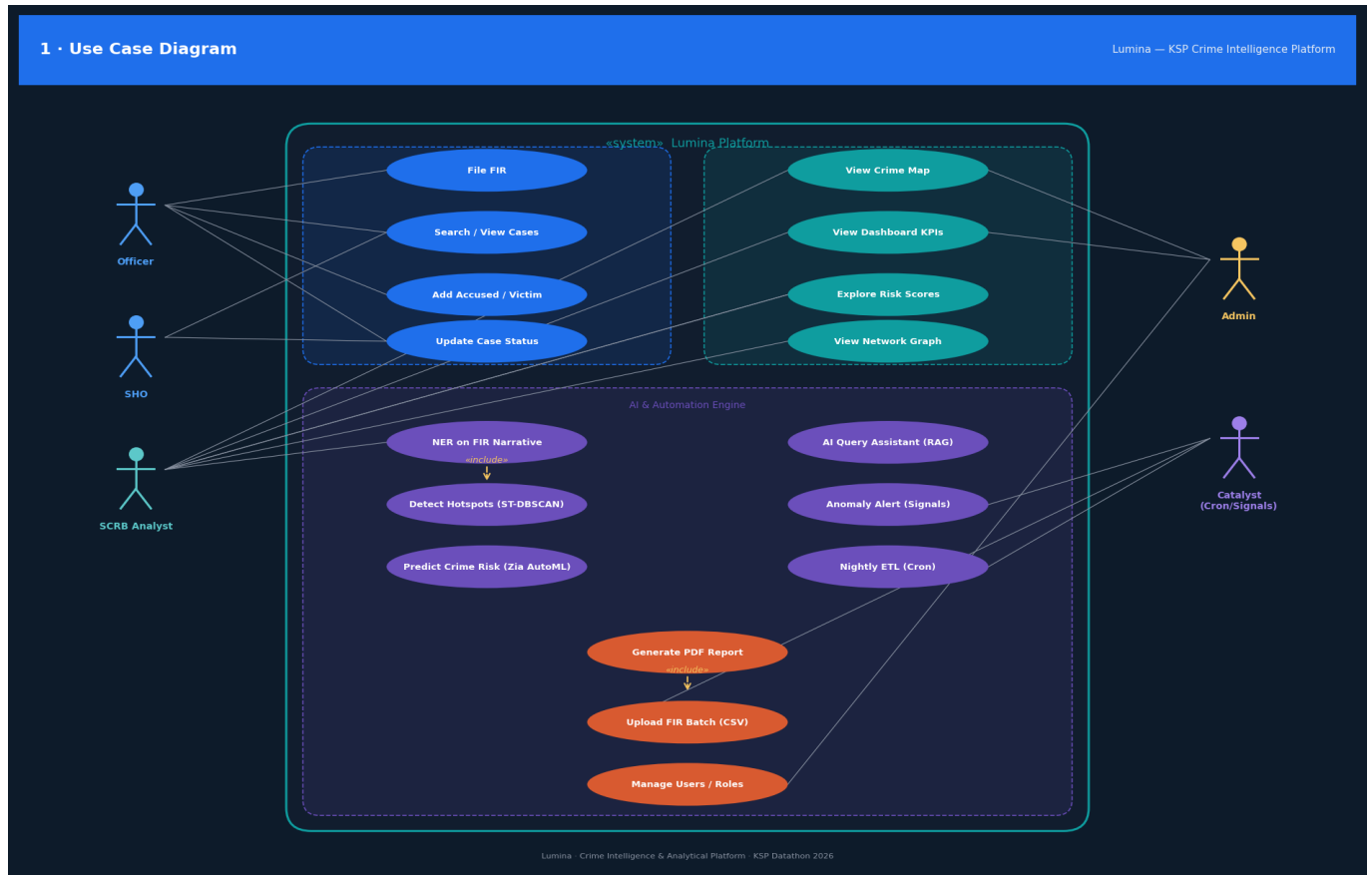
Karnataka State Police · KSP Datathon 2026 · Hack2Skill Track 2

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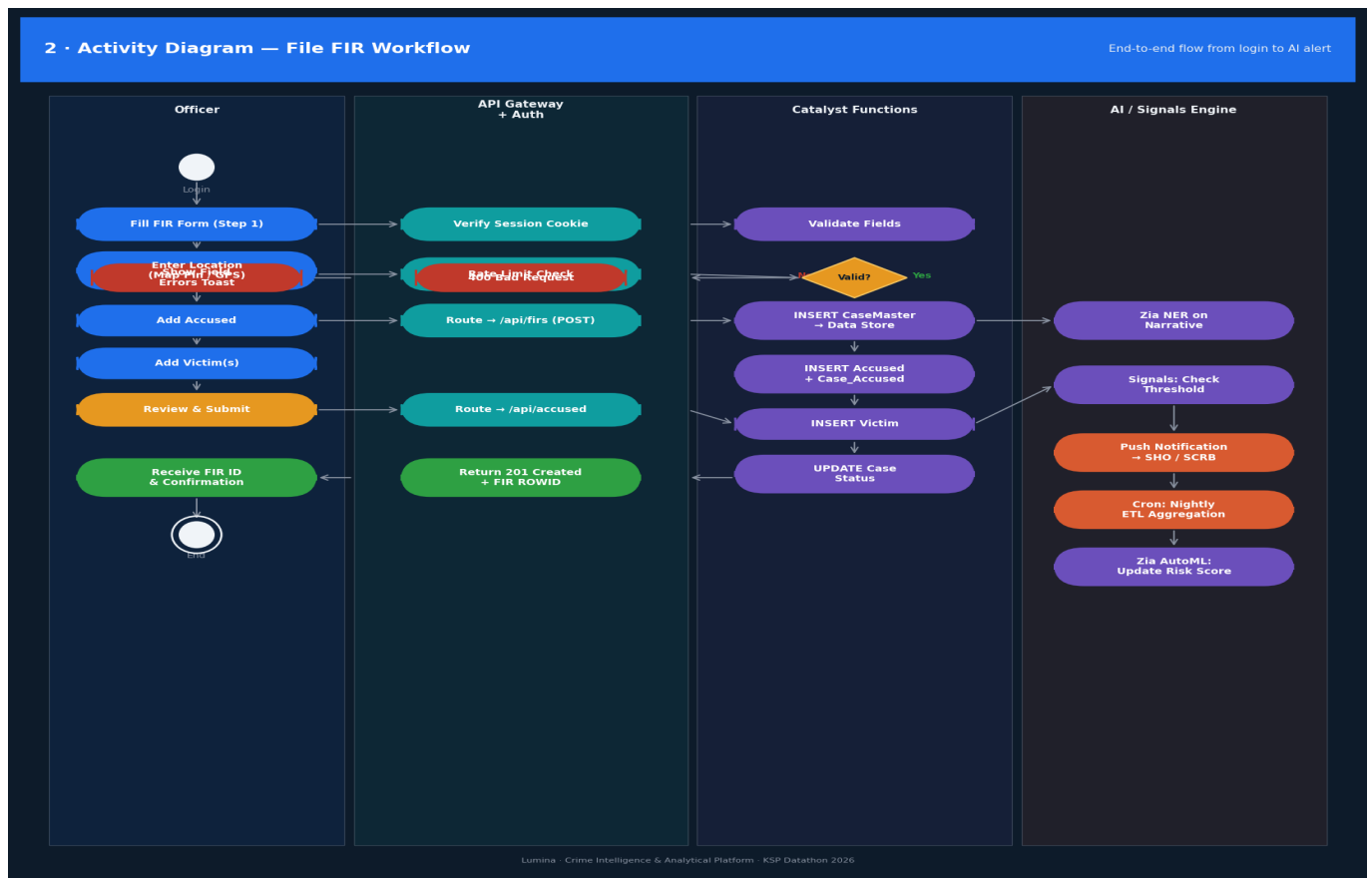
## 1 · Use Case Diagram

Identifies all system actors (Officer, SHO, SCRB Analyst, Admin, Catalyst automation) and the use cases they interact with across the Operational Module, Intelligence/Analytics layer, and AI Engine.



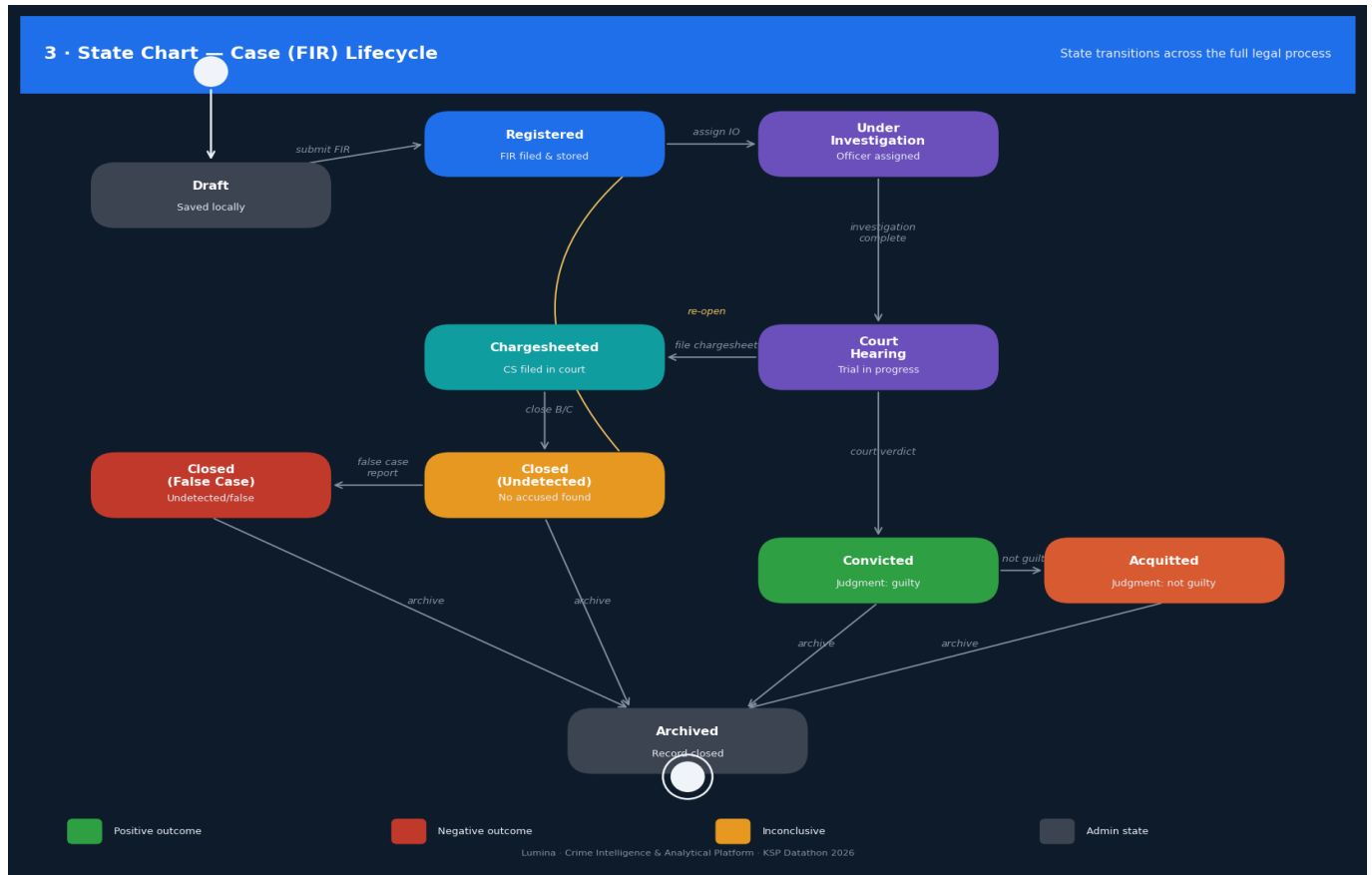
## 2 · Activity Diagram — File FIR Workflow

Swimlane flow across Officer, API Gateway, Catalyst Serverless Functions, and the AI/Signals engine. Covers login → form fill → validation → Data Store insert → NER trigger → Push Notification.



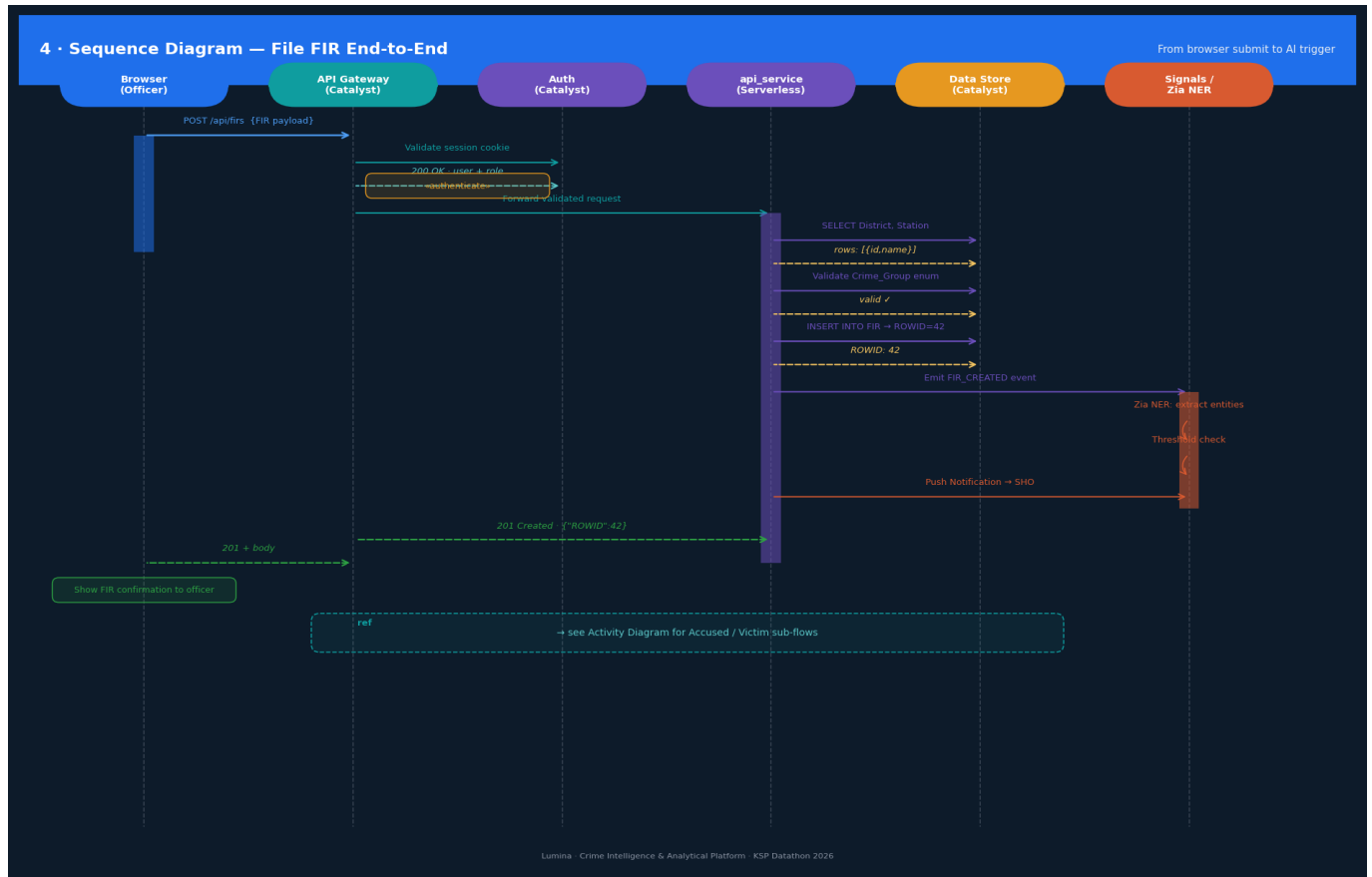
### 3 · State Chart — Case (FIR) Lifecycle

Models every legal state a case can occupy — Draft, Registered, Under Investigation, Chargesheeted, Court Hearing, Convicted, Acquitted, Closed (false/undetected), and Archived — with guard transitions.

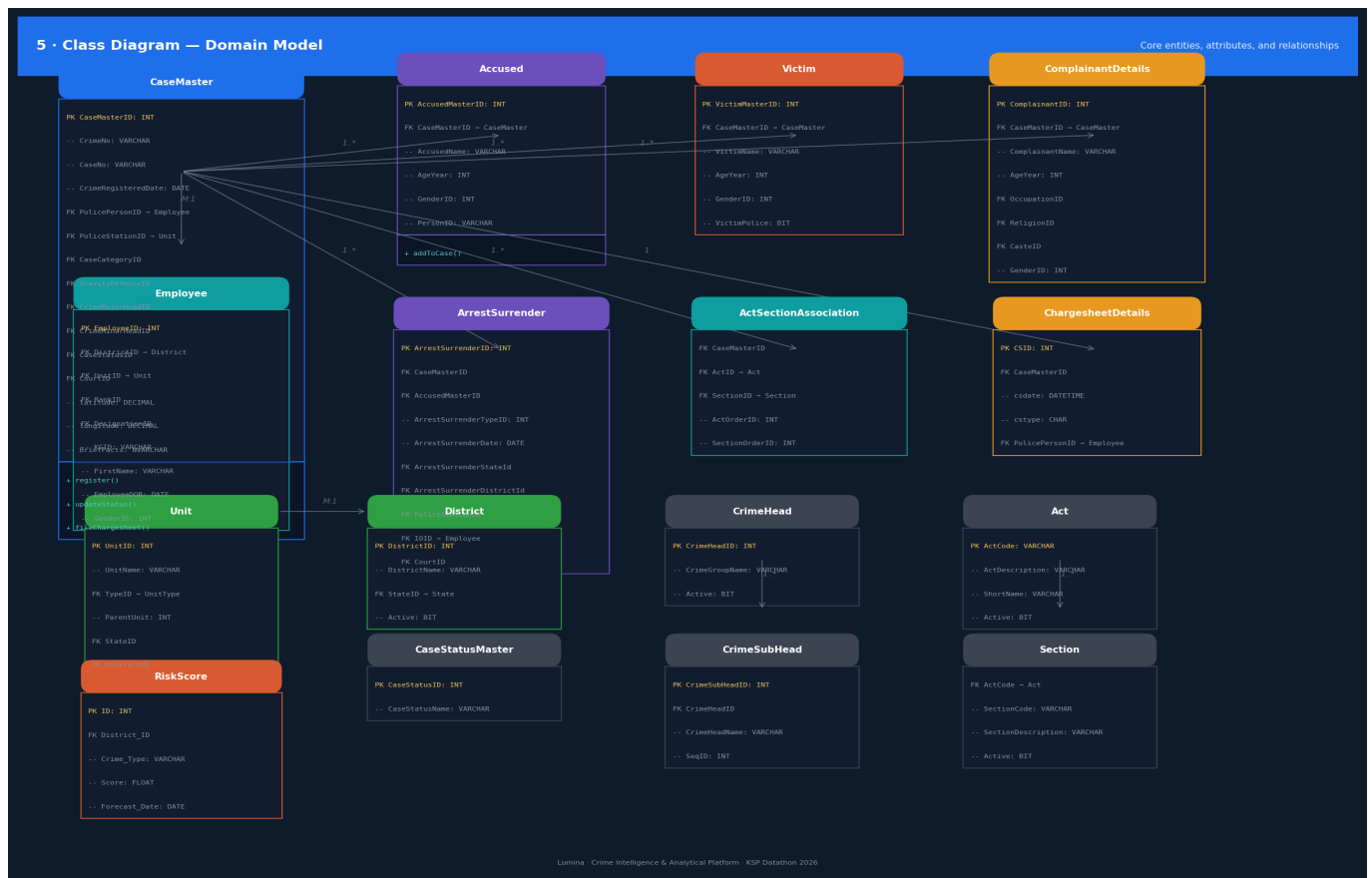


## 4 · Sequence Diagram — File FIR End-to-End

Precise inter-object message flow: Browser → API Gateway → Catalyst Auth → api\_service Serverless → Data Store → Signals/Zia NER. Includes activation bars, return messages, and a ref to sub-flows.



## 5 · Class Diagram — Domain Model



6 · Component Diagram — Lumina Architecture

Five-layer component view mapping every Catalyst service to its role: Ingestion, Storage, AI/ML, Backend Services, and Frontend. Includes Neo4j on AppSail and QuickML RAG pipeline.



## 7 · Deployment Diagram — Zoho Catalyst Cloud

Physical deployment topology showing execution environments (Web Client Hosting, Serverless, AppSail), managed services (Data Store, Stratus, Cache, Auth, Zia, SmartBrowz), and CI/CD via Catalyst Pipelines.

